

Quantum Bilisim Vers 4

$$i) [J_1, J_2, J_{1z}] = [J_{1x} J_{2x}, J_{1z}] + [J_{1y} J_{2y}, J_{1z}] + [J_{1z} J_{2z}, J_{1z}]$$

$$\Rightarrow [J_{1x} J_{1z}] J_{2x} + [J_{1y} J_{1z}] J_{2y} + [J_{1z} J_{1z}] J_{2z} - J_{1y} J_{2x} + J_{1x} J_{2y}$$

$$ii) [J_1 J_2, J_{2z}] = [J_{1x} J_{2x}, J_{2z}] + [J_{1y} J_{2y}, J_{2z}] + 0$$

$$\Rightarrow J_{1x} [J_{2x}, J_{2z}] + J_{1y} [J_{2y}, J_{2z}] = -J_{1x} J_{2y} + J_{1y} J_{2x}$$

$$[J_1, J_2, J_z] = 0$$

$$\{J^2, J_z, J_1^2, J_2^2, \hat{H}\}$$

$\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $J \quad \quad m \quad \quad J_1 \quad \quad J_2$

$$J^2 |J, m, J_1, J_2\rangle = \hbar^2 J(J+1) |J, m, J_1, J_2\rangle$$

$$J_z |J, m, J_1, J_2\rangle = m\hbar |J, m, J_1, J_2\rangle$$

$$J_1^2 |J, m, J_1, J_2\rangle = \hbar^2 J_1(J_1+1) |J, m, J_1, J_2\rangle$$

$$J_2^2 |J, m, J_1, J_2\rangle = \hbar^2 J_2(J_2+1) |J, m, J_1, J_2\rangle$$

$$|J_1, J_2, m_1, m_2\rangle = \text{Bağımsız Baz}$$

$$|J_1, m_1, J_1, J_2\rangle = \text{Bağımsız Baz}$$

$\downarrow$
 Basis
 linear
 indep
 $\downarrow$
 core

$$|\langle m_1, m_2 | J, m \rangle|^2$$

$$\Rightarrow J_z |J, m\rangle = \sum_{m_1=-J_1}^{J_1} \sum_{m_2=-J_2}^{J_2} \langle m_1, m_2 | J, m \rangle J_z |m_1, m_2\rangle$$

$$\downarrow$$

$J_{1z} + J_{2z}$
 $\downarrow \quad \quad \downarrow$
 $m_1 \quad \quad m_2$

$$\Rightarrow \hbar m |J, m\rangle = \sum_{m_1} \sum_{m_2} \langle m_1, m_2 | J, m \rangle \hbar (m_1 + m_2) |m_1, m_2\rangle$$

$$\hbar \sum_{m_1} \sum_{m_2} \langle m_1, m_2 | J, m \rangle |m_1, m_2\rangle = \hbar \sum_{m_1} \sum_{m_2} \langle m_1, m_2 | J, m \rangle (m_1 + m_2) |m_1, m_2\rangle$$

$$\sum_{m_1} \sum_{m_2} \langle m_1, m_2 | J, m \rangle \hbar [m - m_1 - m_2] |m_1, m_2\rangle = 0 \quad |m_1, m_2\rangle = 0$$

$$\langle m_1, m_2 | J, m \rangle [m - m_1 - m_2] = 0 \quad m = m_1 + m_2$$

$$m = m_1 + m_2$$

$$m = -J_1 \dots + J_1$$

$$m_1 = -J_1 \dots + J_1$$

$$m_2 = -J_2 \dots + J_2$$

$$J_{max} = J_1 + J_2 \quad m = m_1 + m_2 \quad m_1 = -J_1 \dots J_1$$
$$m = -J_1 \dots + J_1 \quad m_2 = -J_2 \dots J_2$$
$$\Rightarrow \sum_{J=J_{min}}^{J_{max}} (2J+1) = (2J_1+1)(2J_2+1) \quad J_1, m_1 = \overbrace{-J_1, -J_1+1, \dots, J_1}^{2J_1+1}$$
$$J_2, m_2 = -J_2, -J_2+1, \dots, J_2$$

$$J_{min} \quad m = -J, -J+1, \dots, +J$$
$$(2J+1)$$